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WORK VEHICLE

Abstract

A work vehicle includes a traveling machine body, an in-vehicle operation device that is installed in the traveling machine body and that operates the traveling machine body, and a remote operation device that is provided separately from the traveling machine body and that operates the traveling machine body. The work vehicle has a first operation mode for operating the traveling machine body using the remote operation device and a second operation mode for operating the traveling machine body using the in-vehicle operation device.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a work vehicle.

BACKGROUND ART

[0002] A mowing machine that includes a traveling machine body that performs mowing traveling and can be manually operated using a transmitter (remote operation device) is known as a conventional technique (for example, see Patent Document 1).

PRIOR ART DOCUMENT

Patent Document

[0003] Patent Document 1: Japanese Unexamined Patent Application Publication No. 2019-106941

SUMMARY OF INVENTION

Technical Problem

[0004] In the configuration in Patent Document 1, when the remote operation device is normal, the traveling machine body is operated (particularly, moved to travel) by using the remote operation device so as to be easily moved. However, for example, when a failure or a shortage of power occurs in the remote operation device, the traveling machine body may not be operated by the remote operation device, and therefore, the traveling machine body may not be moved. In this case, in order to move the traveling machine body, for example, it is necessary to prepare a device different from the traveling machine body and lift the traveling machine body, and therefore, it is difficult to move the traveling machine body.

[0005] The present invention has been made to solve the above problem, and an object of the present invention is to provide a technique capable of easily moving a traveling machine body in both a state in which the traveling machine body may be operated by a remote operation device and a state in which the traveling machine body may not be operated by the remote operation device.

Solution to Problem

[0006] According to an aspect of the present invention, a work vehicle includes a traveling machine body, an in-vehicle operation device that is installed in the traveling machine body and that operates the traveling machine body, and a remote operation device that is provided separately from the traveling machine body and that operates the traveling machine body. The work vehicle has a first operation mode for operating the traveling machine body using the remote operation device and a second operation mode for operating the traveling machine body using the in-vehicle operation device.

Advantageous Effects of Invention

[0007] With the configuration described above, the traveling machine body can be easily moved in both a state in which the traveling machine body may be operated by the remote operation device and a state in which the traveling machine body may not be operated by the remote operation device.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a diagram schematically illustrating a configuration of a work vehicle according to an embodiment of the present invention.

[0009] FIG. 2 is a block diagram schematically illustrating a configuration of a work machine lifting device installed in a traveling machine body included in the work vehicle.

[0010] FIG. 3 is a plan view illustrating a configuration of a remote operation device for the work vehicle.

[0011] FIG. **4** is a perspective view illustrating a configuration of an in-vehicle operation device installed in the traveling machine body.

[0012] FIG. **5** is a diagram illustrating a configuration of a switching manipulator installed in the in-vehicle operation device.

[0013] FIG. **6** is a diagram illustrating a configuration of the switching manipulator.

[0014] FIG. **7** is a block diagram schematically illustrating a configuration of a control system of the work vehicle.

[0015] FIG. **8** is an explanatory view illustrating an operation of the traveling machine body in a first operation mode of the work vehicle.

[0016] FIG. **9** is an explanatory view illustrating an operation of the traveling machine body in a second operation mode of the work vehicle.

[0017] FIG. **10** is a flowchart of a flow when an operation instruction is issued by the remote operation device in the second operation mode.

[0018] FIG. **11** is a graph illustrating a change in a target speed of the traveling machine body in the second operation mode with respect to an operation amount of a forward movement operation button of the in-vehicle operation device.

[0019] FIG. **12** is a flowchart of a flow of a setting of switching from the first operation mode to the second operation mode.

[0020] FIG. **13** is a flowchart of a flow of notification by a notifier installed in the traveling machine body.

DESCRIPTION OF EMBODIMENTS

[0021] Embodiments of the present invention will be described below with reference to the drawings.

1. SCHEMATIC CONFIGURATION OF WORK VEHICLE

[0022] FIG. **1** is a diagram schematically illustrating a configuration of a work vehicle **1** according to an embodiment of the present invention. The work vehicle **1** is used to perform work, such as agricultural work and construction work. The work vehicle **1** includes a vehicle main body **10** and a remote operation device **200**. The vehicle main body **10** includes a traveling machine body **11** that travels on the ground, and a work machine **12** that is coupled to the traveling machine body **11**. That is, the work vehicle **1** includes the traveling machine body **11** and the work machine **12**.

[0023] The remote operation device **200** (also referred to as a remote control device) is provided separately from the vehicle main body **10** and enables an operator at a position away from the vehicle main body **10** to operate the vehicle main body **10**, in particular, the traveling machine body **11**. Specifically, the work vehicle **1** includes the remote operation device **200**, and the remote operation device **200** is provided separately from the traveling machine body **11** and operates the traveling machine body **11**. A configuration of the remote operation device **200** will be described below. Furthermore, traveling of the traveling machine body **11** operated by the remote operation device **200**, that is, manually operated by an operator is referred to as manual traveling.

[0024] The work vehicle **1** is configured to be able to travel by automatic traveling in addition to the manual traveling described above. The above-described automatic traveling means that at least steering is autonomously performed along a predetermined route by controlling a device related to traveling (for example, a traveling drive device **111c** described later) by a control device **113** (see FIG. **7**) described later included in the work vehicle **1**. Note that the automatic traveling may be configured such that, in addition to steering, for example, at least one of adjustment of a traveling speed (also simply referred to as a speed) and work by the work machine **12** is autonomously performed. In addition, the work vehicle **1** is not limited to the above-described configuration, and for example, may be capable of manual traveling and incapable of automatic traveling.

[0025] Here, directions used in the description (in particular, directions related to the vehicle main body **10**) are defined as follows. A direction in which the traveling machine body **11** and the work machine **12** are arranged in parallel is referred to as a front-back direction, a direction in which the

work machine **12** is located when viewed from the traveling machine body **11** is referred to as “back”, and an opposite direction is referred to as “front”. Moreover, a right side and a left side from the back side toward the front side are defined as “right” and “left”, respectively. Furthermore, a gravity direction perpendicular to the front-back direction and the right-left direction is defined as an up-down direction, and an upstream side and a downstream side in the gravity direction are defined as “up” and “down”, respectively. In the drawings, the front side, the back side, the right side, the left side, the upside, and the downside are denoted by the symbols “F”, “B”, “R”, “L”, “U”, and “D”, respectively, as necessary. Note that, these directions are names used merely for the explanation and are not intended to limit actual positional relations or directions.

[0026] Note that, in this embodiment, the work machine **12** is disposed behind the traveling machine body **11**, but the present invention can also be applied to a work vehicle **1** in which the work machine **12** is disposed in front of the traveling machine body **11**.

[0027] The traveling machine body **11** includes a machine body portion **111** and a traveling portion **112** disposed below the machine body portion **111**. The machine body portion **111** includes an outer cover **111a**, an in-vehicle operation device **111b** and the traveling drive device **111c** which are disposed on an inner front side covered by the outer cover **111a**, and a work machine drive device **111d** and a work machine lifting device **111e** which are disposed on an inner back side covered by the outer cover **111a**. Specifically, the work vehicle **1** includes the in-vehicle operation device **111b**, and the in-vehicle operation device **111b** is installed in the traveling machine body **11**.

[0028] The in-vehicle operation device **111b** enables the operator to operate the traveling machine body **11**. That is, the in-vehicle operation device **111b** operates the traveling machine body **11**. In addition, the in-vehicle operation device **111b** enables various settings of the work vehicle **1** (the traveling machine body **11**, the work machine **12**, and the remote operation device **200**). A configuration of the in-vehicle operation device **111b** will be described below.

[0029] Note that, in this embodiment, the work vehicle **1** has a first operation mode and a second operation mode as operation modes. In the first operation mode, the traveling machine body **11** is operated (for example, the traveling machine body **11** is moved to travel) using the remote operation device **200**. In the second operation mode, the traveling machine body **11** is operated (for example, the traveling machine body **11** is moved to travel) using the in-vehicle operation device **111b**. The operations of the traveling machine body **11** in the individual operation modes will be described in detail below.

[0030] The traveling drive device **111c** includes a drive source and a power transmission portion (not illustrated) that transmits power from the drive source to the traveling portion **112**. In this embodiment, the drive source included in the traveling drive device **111c** is an electric motor. Note that the drive source included in the traveling drive device **111c** may be other than the electric motor, and may be, for example, an engine.

[0031] The work machine drive device **111d** includes a drive source and a power take off (PTO) power transmission portion (not illustrated) that can transmit power from the drive source to an outside of the traveling machine body **11**. In this embodiment, the drive source included in the work machine drive device **111d** is an electric motor. Note that the drive source included in the work machine drive device **111d** may be other than the electric motor, and may be, for example, an engine.

[0032] Furthermore, in this embodiment, although the drive source included in the traveling drive device **111c** and the drive source included in the work machine drive device **111d** are different from each other, a drive source (electric motor) may be shared by the traveling drive device **111c** and the work machine drive device **111d**. Note that a configuration of the work machine lifting device **111e** will be described later.

[0033] A notifier **111f** is further disposed inside the outer cover **111a**, in particular, on an upper side of the inside. That is, the work vehicle **1** includes the notifier **111f**. In this embodiment, the notifier **111f** is a speaker. Note that the notifier **111f** is not limited to the speaker, and may be, for example,

a buzzer or a visual notifier. Examples of the visual notifier include a lamp using an incandescent lamp as a light source, a lamp using a light emitting diode (LED) as a light source, a liquid crystal display device, and an organic EL display device. Furthermore, an arrangement position of the notifier **111f** is not limited to the above, and may be, for example, outside the outer cover **111a**. Note that a time point (timing) when the notifier **111f** performs notification will be described later. [0034] In addition, a battery that supplies electric power to the electric motor, a power electronics device (both not illustrated), and the like are disposed inside the outer cover **111a**. As an example, a light **111g**, a positioning antenna **111h**, an alarm lamp **111j**, and the like are disposed outside the outer cover **111a**.

[0035] The traveling portion **112** supports the machine body portion **111** in a traveling available manner. Specifically, the traveling portion **112** includes a pair of right and left crawlers **112a**. The right and left crawlers **112a** include respective track frames **112b** extending in the front-back direction. The individual track frames **112b** are attached to a lower surface of the machine body portion **111**. The track frames **112b** have respective drive sprockets **112c** arranged as drive wheels at respective front ends of the track frames **112b**. Power from the electric motor is transmitted to the drive sprockets **112c** via a power transmitter included in the traveling drive device **111c**. Driven sprockets **112d** are arranged as driven wheels at respective back ends of the track frames **112b**. The driven sprockets **112d** are rotatably supported by the track frames **112b**. In each of the track frames **112b**, a plurality of rolling wheels **112e** are rotatably supported between the drive sprocket **112c** and the driven sprocket **112d**. To form each of the crawlers **112a**, a crawler belt **112f** is wound around the drive sprocket **112c**, the driven sprocket **112d**, and the plurality of rolling wheels **112e**.

[0036] The right and left crawlers **112a** are driven by different electric motors included in the traveling drive device **111c**. When the pair of right and left crawlers **112a** are simultaneously driven in the same direction, the traveling portion **112** travels straight forward or backward. Whether the traveling portion **112** travels straight forward or backward depends on a rotation direction of the electric motors. For example, when the pair of right and left crawlers **112a** are independently driven, the traveling portion **112** turns right or turns left.

[0037] Note that, in this embodiment, each of the crawlers **112a** is configured such that one drive wheel (drive sprocket **112c**) and one driven wheel (driven sprocket **112d**) are arranged side by side in the front-back direction and the crawler belt **112f** is wound around the drive wheel and the driven wheel. However, another configuration may be employed. For example, the crawlers may be of a type in which a crawler belt is wound around one drive wheel and two driven wheels in a triangular shape. Moreover, in this embodiment, the traveling portion **112** is of a crawler type, but may be of a type other than the crawler type, for example, a wheel type.

[0038] The work machine **12** is attached to the traveling machine body **11** via a hitch portion **13** in a lifting available manner. The hitch portion **13** includes the work machine drive device **111d**. The work machine **12** is attached to the hitch portion **13** so as to be replaceable. That is, various types of the work machine **12** may be attached. In FIG. 1, the work machine **12** is a tiller. The work machine **12** may be, for example, a plow, a ridging device, a fertilizing device, an agricultural chemical spraying device, a harvesting device, a reaping device, a snow removal device, or the like, in addition to a tiller.

[0039] Here, a configuration of the work machine lifting device **111e** will be described with reference to FIG. 2. FIG. 2 is a block diagram schematically illustrating a configuration of the work machine lifting device **111e**. The work machine lifting device **111e** includes a hydraulic pump **111e1**, a directional control valve **111e2**, and a lifting cylinder **111e3**.

[0040] In this embodiment, the hydraulic pump **111e1** is an electric hydraulic pump that incorporates an electric motor (not illustrated) and is driven by the electric motor. Note that, in this embodiment, the hydraulic pump **111e1** is driven when an instruction for lifting or lowering the work machine **12** is issued. Therefore, when the instruction for lifting or lowering the work machine **12** is not issued, the hydraulic pump **111e1** is stopped. Furthermore, the hydraulic pump

111e1 is not limited to an electric pump. For example, the hydraulic pump **111e1** may be configured to have an engine disposed separately from the hydraulic pump **111e1** as a drive source and be driven by power output from the engine.

[0041] The hydraulic pump **111e1** is connected to a hydraulic oil tank (not illustrated) that stores hydraulic oil. When the hydraulic pump **111e1** is driven, hydraulic oil in the hydraulic oil tank is supplied to the lifting cylinder **111e3** via the directional control valve **111e2**. The directional control valve **111e2** controls a flow direction and a flow rate of the hydraulic oil supplied to the lifting cylinder **111e3**.

[0042] The lifting cylinder **111e3** is constituted by a hydraulic cylinder in this embodiment. More specifically, a base end portion of the lifting cylinder **111e3** is coupled to the machine body portion **111**, and a tip end portion thereof is connected to the hitch portion **13**. When the lifting cylinder **111e3** expands or contracts, the hitch portion **13** turns in the up-down direction with respect to the traveling machine body **11**. As described above, the work machine **12** is attached to the hitch portion **13**. Therefore, the work machine **12** can be moved in the up-down direction by expansion and contraction of the lifting cylinder **111e3**. The lifting cylinder **111e3** is not limited to the above-described configuration, and may be, for example, an electric cylinder.

2. CONFIGURATION OF REMOTE OPERATION DEVICE

[0043] A configuration of the remote operation device **200** will be described with reference to FIG. 3. FIG. 3 is a plan view illustrating a configuration of the remote operation device **200**. The remote operation device **200** includes a housing **201**, a power switch **202**, an antenna **203**, operation levers **204**, operation switches **205**, operation knobs **206**, and a display **207**.

[0044] The housing **201** configures a main body portion of the remote operation device **200**. The power switch **202**, the antenna **203**, the operation levers **204**, the operation switches **205**, the operation knobs **206**, and the display **207** described above are individually mounted on appropriate portions of the housing **201**. Note that the arrangement illustrated in FIG. 3 is merely an example and may be changed appropriately.

[0045] The power switch **202** is disposed at a center of a front surface of the housing **201**, and enables turning on and off of the power of the remote operation device **200**. The power switch **202** is, for example, a seesaw switch. Note that a power source of the remote operation device **200** is, for example, a battery, a dry cell, or the like disposed in the housing **201**.

[0046] The antenna **203** protrudes from a side surface (side surface on an upper side in FIG. 3) of the housing **201** and enables wireless communication with the traveling machine body **11**. When the power source of the remote operation device **200** is turned on by the power switch **202**, the remote operation device **200** can perform wireless communication with the traveling machine body **11**. When the power of the remote operation device **200** is turned off by the power switch **202**, the remote operation device **200** may not communicate with the traveling machine body **11**. In this embodiment, when the traveling machine body **11** may not communicate with the remote operation device **200**, the traveling machine body **11** automatically stops traveling. That is, the power switch **202** has a function as an emergency stop switch of the traveling machine body **11**.

[0047] Note that the emergency stop switch may be provided separately from the power switch **202**.

[0048] The operation levers **204** include a first operation lever **204a** and a second operation lever **204b** arranged side by side with the power switch **202** interposed therebetween. The first operation lever **204a** (the left-side lever illustrated in FIG. 3) can be pulled down at least in two directions (F1-B1 direction and L1-R1 direction) orthogonal to each other, which are indicated by broken arrows in FIG. 3. Furthermore, the first operation lever **204a** may be pulled down in one direction in the L1-R1 direction while being pulled down in one direction in the F1-B1 direction. Note that the second operation lever **204b** (the right-side lever illustrated in FIG. 3) can also be pulled down in the same directions as the first operation lever **204a**. Functions of the first operation lever **204a** and the second operation lever **204b** will be described below.

[0049] The operation switches **205** include a first operation switch **205a** and a second operation switch **205b**. The first operation switch **205a** is disposed on a side surface (a side surface on an upper right side in FIG. 3) of the housing **201**. The first operation switch **205a** is, for example, a momentary switch that can be pressed down. The second operation switch **205b** is disposed on a side surface (a side surface on an upper left side in FIG. 3) of the housing **201**. The second operation switch **205b** is a toggle switch, for example. Functions of the first operation switch **205a** and the second operation switch **205b** will be described later.

[0050] The operation knobs **206** are disposed on the front surface (an upper side of the front surface in FIG. 3) of the housing **201**, and enable adjustment of a maximum speed V_{max} (see FIG. 11) of the traveling machine body **11**. Specifically, two operation knobs **206** are disposed. One of the two operation knobs **206** enables adjustment of the maximum speed V_{max} of the traveling machine body **11** during straight traveling. The other of the two operation knobs **206** enables adjustment of the maximum speed V_{max} of the traveling machine body **11** during turning traveling.

[0051] The display **207** is disposed on the front surface (a lower side of the front surface in FIG. 3) of the housing **201** and displays various types of information as notifications for an operator. The various types of information include, for example, a display of a speed V (see FIG. 11) of the traveling machine body **11**. The display **207** is, for example, a liquid crystal display device, an organic EL display device, or the like.

3. CONFIGURATION OF IN-VEHICLE OPERATION DEVICE

[0052] A configuration of the in-vehicle operation device **111b** will be described with reference to FIG. 4. FIG. 4 is a perspective view illustrating a configuration of the in-vehicle operation device **111b**. Note that, in FIG. 4, for convenience, a portion of the outer cover **111a** located over the in-vehicle operation device **111b** is not illustrated.

[0053] As described above, the in-vehicle operation device **111b** is disposed on an inner front side covered by the outer cover **111a** (see also FIG. 1). That is, the in-vehicle operation device **111b** is disposed on a front side of the traveling machine body **11**. That is, the in-vehicle operation device **111b** is disposed on one side (the front side in this embodiment) in one direction (the front-back direction in this embodiment) included in the horizontal direction in the traveling machine body **11**. Note that an arrangement position of the in-vehicle operation device **111b** is not limited to the above. For example, the in-vehicle operation device **111b** may be disposed on the back side of the traveling machine body **11**, or may be disposed on any one of the right side and the left side of the traveling machine body **11**. Specifically, the one direction only needs to be included in the horizontal direction, and may be the right-left direction.

[0054] The in-vehicle operation device **111b** includes a light switch **111b1**, a traveling manipulator **111b2**, and a setting manipulator **111b3**.

[0055] The light switch **111b1** is disposed on an upper back side of the in-vehicle operation device **111b**, and is capable of switching a light **111g** (see FIG. 1) between on and off. The light switch **111b1** is, for example, a seesaw switch. The traveling manipulator **111b2** is disposed in an upper portion of the in-vehicle operation device **111b**. A configuration of the traveling manipulator **111b2** will be described later.

[0056] The setting manipulator **111b3** is disposed on the front side of the in-vehicle operation device **111b**, and is constituted by a touch panel. The setting manipulator **111b3** displays various types of information of the work vehicle **1** (the traveling machine body **11**, the work machine **12**, and the remote operation device **200**) and enables various types of settings of the work vehicle **1**. That is, a screen (image) for displaying the various types of information or a screen (image) for performing the various types of setting are displayed on the setting manipulator **111b3**.

[0057] Note that the setting manipulator **111b3** is not limited to the above-described configuration. For example, the setting manipulator **111b3** may be configured by a switch, a dial, a lever, or the like. Furthermore, in order to display various types of information, a display different from the

setting manipulator **111b3** may be installed in the in-vehicle operation device **111b**.

[0058] The various settings include a setting for switching operation modes (a first operation mode and a second operation mode) of the traveling machine body **11**. The operation mode switching setting is performed by operating the switching manipulator **111b4** (also referred to as a switching operation screen) displayed on the setting manipulator **111b3**. In this embodiment, the switching manipulator **111b4** includes a screen (image) displayed on the setting manipulator **111b3**.

Furthermore, as described above, the setting manipulator **111b3** is disposed in the in-vehicle operation device **111b**, and the in-vehicle operation device **111b** is installed in the traveling machine body **11**. Specifically, the work vehicle **1** includes the switching manipulator **111b4**, and the switching manipulator **111b4** is installed in the traveling machine body **11**. Hereinafter, a configuration (screen configuration) of the switching manipulator **111b4** will be described.

[0059] FIGS. **5** and **6** are diagrams illustrating a configuration of the switching manipulator **111b4**. Note that, in FIGS. **5** and **6**, in a first operation mode display icon **IC1a** and a second operation mode display icon **IC2a** which will be described later, white filling (not filling) indicates turn-off and black filling indicates turn-on (or blinking).

[0060] The switching manipulator **111b4** includes a first operation mode setting icon **IC1** and a second operation mode setting icon **IC2**. Furthermore, the first operation mode setting icon **IC1** includes the first operation mode display icon **IC1a**, and the second operation mode setting icon **IC2** includes the second operation mode display icon **IC2a**.

[0061] The first operation mode setting icon **IC1** enables a setting of the first operation mode. The first operation mode display icon **IC1a** is turned on (or blinks) when the first operation mode is set (in the work vehicle **1**) (see FIG. **5** in particular). On the other hand, the first operation mode display icon **IC1a** is turned off when the first operation mode is not set, that is, when the second operation mode is set (in the work vehicle **1**) (see FIG. **6** in particular).

[0062] The second operation mode setting icon **IC2** enables a setting of the second operation mode. The second operation mode display icon **IC2a** is turned on (or blinks) when the second operation mode is set (in the work vehicle **1**) (see FIG. **6** in particular). On the other hand, the second operation mode display icon **IC2a** is turned off when the second operation mode is not set, that is, when the first operation mode is set (in the work vehicle **2**) (see FIG. **5** in particular).

[0063] In this embodiment, the switching manipulator **111b4** for performing a switching setting of the operation mode includes a screen displayed on the setting manipulator **111b3**, but is not limited to this configuration. For example, the switching manipulator **111b4** may be disposed separately from the setting manipulator **111b3**. Furthermore, the switching manipulator **111b4** may be a switch, a dial, a lever, or the like.

4. CONFIGURATION OF CONTROL SYSTEM OF WORK VEHICLE

[0064] A configuration of a control system of the work vehicle **1** will be described with reference to FIG. **7**. FIG. **7** is a block diagram schematically illustrating a configuration of a control system of the work vehicle **1**. Note that, in FIG. **7**, components required to describe the features of this embodiment are illustrated, and common components are not illustrated.

[0065] The traveling machine body **11** includes the control device **113**. In particular, the control device **113** is installed in the machine body portion **111** of the traveling machine body **11**. The control device **113** is, for example, a computer device that is configured to include an arithmetic device, an input/output portion, and a storage **113a**. The arithmetic device is, for example, a processor or a microprocessor. The storage **113a** is a main storage device, such as a read only memory (ROM) or a random access memory (RAM). The storage **113a** may further include an auxiliary storage device, such as a hard disk drive (HDD) or a solid state drive (SSD). The storage **113a** stores various types of program, data, and the like. The arithmetic device reads out the various types of program from the storage **113a** and executes arithmetic processing in accordance with the programs. The programs stored in the storage **113a** may be provided by, for example, a computer-readable non-volatile recording medium. As another example, the programs may be provided from

a program providing server via a communication line, such as the Internet.

[0066] The control device **113** may be operated as an operation mode switcher **113b**, a traveling controller **113c**, a work machine controller **113d**, and a notification controller **113e** by cooperation of the above-described hardware and software. The control device **113** may be configured by a single piece of hardware, or may be configured by a plurality of pieces of hardware that can communicate with each other.

[0067] Note that, as described above, the functional portions **113b** to **113e** included in the control device **113** may be realized by causing the arithmetic device to execute arithmetic processing in accordance with the programs, that is, by software, but may be realized by another method. At least one of the functional portions **113b** to **113e** may be implemented using, for example, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or the like. That is, at least one of the functional portions **113b** to **113e** may be realized by hardware using a dedicated IC or the like. Moreover, at least one of the functional portions **113b** to **113e** may be realized by using both software and hardware. In addition, each of the functional portions **113b** to **113e** has a conceptual configuration. Therefore, a function executed by one component may be distributed to a plurality of components, or functions of a plurality of components may be integrated into one component.

[0068] The operation mode switcher **113b** controls switching between a first operation mode and a second operation mode based on a switching instruction (setting instruction) issued by the switching manipulator **111b4**. A flow of the operation mode switching setting will be described later.

[0069] The traveling controller **113c** controls the traveling drive device **111c** (see FIG. 1) in accordance with whether the work vehicle **1** is set to the first operation mode or the second operation mode. Functions of the traveling controller **113c** in the individual operation modes will be described later.

[0070] The work machine controller **113d** controls the work machine drive device **111d** (see FIG. 1) and the work machine lifting device **111e** (see FIGS. 1 and 2) in accordance with whether the work vehicle **1** is set to the first operation mode or the second operation mode. Functions of the work machine controller **113d** in the individual operation modes will be described later.

[0071] When the work vehicle **1** is set to the second operation mode, the notification controller **113e** drives (sounds) the notifier **111f** (a speaker in this embodiment) based on a traveling direction of the traveling machine body **11**.

[0072] A positioning communication unit **114**, a communication processor **115**, and a sensor **116** are connected to the control device **113**. Note that the traveling manipulator **111b2** and the setting manipulator **111b3** are also connected to the control device **113**.

[0073] The positioning communication unit **114** includes the positioning antenna **111h** (see FIG. 1) and acquires a position of the vehicle main body **10** as information on latitudes and longitudes, for example, by using a positioning signal received by the positioning antenna **111h** from a positioning satellite. For example, the positioning communication unit **114** receives a positioning signal from a reference station, not illustrated, by an appropriate method, and then performs positioning by using a known real time kinematic GNSS (RTK-GNSS) method. The positioning communication unit **114** outputs positional information of the vehicle main body **10** to the control device **113**. The positioning communication unit **114** may perform positioning by using another method, for example, a differential GNSS (DGNSS) method. Furthermore, the work vehicle **1** may include, for example, a quantum compass capable of performing positioning, instead of or in addition to the positioning communication unit **114**.

[0074] The communication processor **115** performs communication with the remote operation device **200** via a communication antenna **115a**. The communication antenna **115a** is used to perform wireless communication with the remote operation device **200**. As the wireless communication, wireless Local Area Network (LAN), such as Wi-Fi (registered trademark), may

be used.

[0075] The sensor **116** detects information related to the vehicle main body **10** and outputs the detected information to the control device **113**. In this embodiment, the sensor **116** includes a plurality of types of sensor. Each of the plurality of types of sensor is connected to the control device **113** so as to be able to input a signal. Examples of the plurality of types of sensor include an inertial measurement device, an obstacle sensor, a speed sensor, and a lifting position sensor.

[0076] The inertial measurement device includes a tri-axial angular velocity sensor and a tri-directional acceleration sensor and can measure an attitude of the vehicle main body **10**. The obstacle sensor detects an obstacle existing around the vehicle main body **10**, and may be, for example, an ultrasonic sensor, a camera, a radar, or light detection and ranging (LiDAR). The speed sensor detects a speed *V* (see FIG. **11**) of the traveling machine body **11**. The lifting position sensor detects a lifting position (height from the ground) of the work machine **12**.

[0077] Note that the positioning communication unit **114** and some of the sensors **116** (for example, the inertial measurement device) are used when the work vehicle **1** automatically travels. That is, the positioning communication unit **114** and some of the sensors **116** are not essential for the work vehicle **1** to travel based on an instruction issued from the remote operation device **200** or the in-vehicle operation device **111b**.

5. OPERATION OF TRAVELING MACHINE BODY IN FIRST OPERATION MODE AND SECOND OPERATION MODE

[0078] Operations of the traveling machine body **11** in the first operation mode and the second operation mode will be described. First, the operation of the traveling machine body **11** in the first operation mode will be described with reference to FIG. **8**.

[0079] FIG. **8** is an explanatory view illustrating the operation of the traveling machine body **11** in the first operation mode.

[0080] In the first operation mode, when the first operation lever **204a** of the remote operation device **200** is tilted in the F1 direction, the traveling machine body **11** can be moved forward. More specifically, a forward movement instruction is output from the remote operation device **200** by tilting the first operation lever **204a** in the F1 direction. The forward movement instruction is transmitted to the control device **113** via the communication antenna **115a** and the communication processor **115** (see FIG. **7**). In the control device **113**, the traveling controller **113c** controls the traveling drive device **111c** (in particular, the electric motor) to set an output (rotational output) of the traveling drive device **111c** to a rotational speed and a direction corresponding to the forward movement instruction. Thus, the traveling portion **112** (see FIG. **1**) is driven, and the traveling machine body **11** moves forward. Note that, as described above, since the work machine **12** is attached to the traveling machine body **11** via the hitch portion **13**, when the traveling machine body **11** moves forward, the work machine **12** also moves forward.

[0081] Similarly, in the first operation mode, when the first operation lever **204a** is tilted in the B1 direction, the traveling machine body **11** can be moved backward. In the first operation mode, when the first operation lever **204a** is tilted in the L1 direction, the traveling machine body **11** can be turned to the left. In the first operation mode, when the first operation lever **204a** is tilted in the R1 direction, the traveling machine body **11** can be turned to the right.

[0082] In the first operation mode, when the second operation lever **204b** of the remote operation device **200** is tilted in the L1 direction, the traveling machine body **11** can be turned to the left. In the first operation mode, when the second operation lever **204b** is tilted in the R1 direction, the traveling machine body **11** can be turned to the right.

[0083] Note that a turning radius (turning amount) obtained when the operation is performed with the same operation amount (operation amount from the neutral position) differs between the turning by the first operation lever **204a** (right turning and left turning) and the turning by the second operation lever **204b** (right turning and left turning). More specifically, when the first operation lever **204a** and the second operation lever **204b** are tilted by the same amount from a

neutral position, the traveling machine body **11** can be turned with a larger turning radius when being operated by the second operation lever **204b** than when being operated by the first operation lever **204a**. In other words, the traveling machine body **11** can be gently turned by the second operation lever **204b**. Note that the configurations of the first operation lever **204a** and the second operation lever **204b** are not limited to the above, and for example, a configuration may be adopted in which the traveling machine body **11** can be gently turned by the first operation lever **204a** relative to the second operation lever **204b**.

[0084] In the first operation mode, when the second operation lever **204b** is tilted in the F1 direction, the work machine **12** can be lifted. In the first operation mode, when the second operation lever **204b** is tilted in the B1 direction, the work machine **12** can be lowered. More specifically, by tilting the second operation lever **204b** in the F1 direction (B1 direction), an instruction to lift (instruction to lower) the work machine **12** is output from the remote operation device **200**. This instruction is transmitted to the control device **113** via the communication antenna **115a** and the communication processor **115**. In the control device **113**, the work machine controller **113d** controls the work machine lifting device **111e** such that the work machine **12** is lifted (lowered). Specifically, the work machine controller **113d** starts driving of the hydraulic pump **111e1** and controls the directional control valve **111e2** to supply hydraulic oil in a direction and at a flow rate corresponding to the instruction to the lifting cylinder **111e3**. As a result, the lifting cylinder **111e3** is extended (contracted) and the work machine **12** is lifted (lowered).

[0085] Furthermore, in the first operation mode, when the first operation switch **205a** is pressed, a plurality of types of setting related to automatic traveling can be performed. The plurality of types of setting include, for example, a start setting and an end setting of the automatic traveling. That is, in the first operation mode, when the first operation switch **205a** is pressed, the automatic traveling can be started or ended.

[0086] In the first operation mode, when the second operation switch **205b** is operated, it is possible to perform switching between a state in which power transmission to the work machine **12** using the above-described PTO power transmission portion is possible and a state in which the power transmission is not possible. That is, the work machine **12** can be driven or stopped. The work machine **12** is driven and stopped by the work machine controller **113d** of the control device **113** controlling the work machine drive device **111d** (in particular, the electric motor as the drive source) based on an operation (instruction) issued by the second operation switch **205b**.

[0087] Therefore, in the first operation mode, the traveling machine body **11** can travel, the work machine lifting device **111e** can be driven (the work machine **12** can be lifted and lowered), and the work machine drive device **111d** can be driven (the work machine **12** can be driven) by operating the remote operation device **200**. That is, in the first operation mode, the traveling machine body **11** can be operated using the remote operation device **200**.

[0088] Next, the operation of the traveling machine body **11** in the second operation mode will be described with reference to FIG. 9. FIG. 9 is an explanatory view illustrating the operation of the traveling machine body **11** in the second operation mode. First, a configuration of the traveling manipulator **111b2** of the in-vehicle operation device **111b** for operating the traveling machine body **11** in the second operation mode will be described.

[0089] In this embodiment, the traveling manipulator **111b2** includes two operation members OP. The two operation members OP are arranged side by side in the right-left direction. Here, the operation member OP positioned on the right side is referred to as a first operation member OP1, and the operation member OP positioned on the left side is referred to as a second operation member OP2. Each of the operation members OP includes two operation buttons OB. The operation buttons OB are, for example, momentary push buttons.

[0090] In the first operation member OP1, the two operation buttons OB are arranged side by side in the front-back direction. Here, in the first operation member OP1, of the two operation buttons OB, the operation button OB positioned on the front side is referred to as a forward movement

operation button OB-F, and the operation button OB positioned on the back side is referred to as a backward movement operation button OB-B.

[0091] In the second operation member OP2, two operation buttons OB are arranged side by side in the right-left direction. Here, in the second operation member OP2, of the two operation buttons OB, the operation button OB positioned on the left side is referred to as a left turning operation button OB-L, and the operation button OB positioned on the right side is referred to as a right turning operation button OB-R.

[0092] Note that the traveling manipulator **111b2** is not limited to the above-described configuration, and may be configured by, for example, a switch, a dial, a lever, or the like.

Alternatively, the traveling manipulator **111b2** may be configured by a touch panel. That is, the traveling manipulator **111b2** may include a screen displayed on the setting manipulator **111b3**.

[0093] In the second operation mode, when the forward movement operation button OB-F is pressed, the traveling machine body **11** can be moved forward. More specifically, when the forward movement operation button OB-F is pressed, a forward movement instruction is output from the in-vehicle operation device **111b** (traveling manipulator **111b2**). The forward movement instruction is transmitted to the control device **113** (see FIG. 7). In the control device **113**, the traveling controller **113c** controls the traveling drive device **111c** (in particular, the electric motor) to set an output (rotational output) of the traveling drive device **111c** to a rotational speed and a direction corresponding to the forward movement instruction. As a result, the traveling portion **112** is driven, and the traveling machine body **11** moves forward. Note that, in the second operation mode, a speed of the traveling machine body **11** is limited. The speed limit of the traveling machine body **11** will be described later.

[0094] Similarly, in the second operation mode, when the backward movement operation button OB-B is pressed, the traveling machine body **11** can be moved backward. In the second operation mode, when the left turning operation button OB-L is pressed, the traveling machine body **11** can be turned left. In the second operation mode, when the right turning operation button OB-R is pressed, the traveling machine body **11** can be turned right. That is, (particularly in the second operation mode), the traveling manipulator **111b2** can issue a traveling instruction to the traveling machine body **11**.

[0095] Note that one of the left turning operation button OB-L and the right turning operation button OB-R may be pressed together with one of the forward movement operation button OB-F and the backward movement operation button OB-B. For example, in the second operation mode, when the left turning operation button OB-L is pressed together with the forward movement operation button OB-F, the traveling machine body **11** can be turned left while being moved forward.

[0096] Therefore, in the second operation mode, the traveling machine body **11** can be caused to travel by operating the in-vehicle operation device **111b** (particularly, the traveling manipulator **111b2**). That is, in the second operation mode, the traveling machine body **11** can be operated using the in-vehicle operation device **111b**.

[0097] According to the above configuration, when the work vehicle **1** is set to the first operation mode in a state where the traveling machine body **11** can be operated by the remote operation device **200**, the traveling machine body **11** can be operated by the remote operation device **200**. Therefore, the operator can easily move the traveling machine body **11** by operating the remote operation device **200** (by setting the work vehicle **1** to the first operation mode). On the other hand, for example, even when the traveling machine body **11** may not be operated by the remote operation device **200** due to a failure, power shortage, or the like of the remote operation device **200**, the traveling machine body **11** can be operated by the in-vehicle operation device **111b** by setting the work vehicle **1** to the second operation mode. Therefore, even in the above case, the operator can easily move the traveling machine body **11** by operating the in-vehicle operation device **111b** (by setting the work vehicle **1** to the second operation mode). As described above, it is

possible to easily move the traveling machine body **11** in both the state in which the operation of the traveling machine body **11** by the remote operation device **200** is possible and the state in which the operation is impossible.

[0098] Note that, in the second operation mode, even when at least one of the first operation lever **204a**, the second operation lever **204b**, the first operation switch **205a**, and the second operation switch **205b** of the remote operation device **200** is operated, the traveling machine body **11** does not move (does not operate). That is, in the second operation mode, the traveling machine body **11** is prohibited from being operated using the remote operation device **200**. More details are as follows. FIG. **10** is a flowchart of a flow when an operation instruction is issued by the remote operation device **200** in the second operation mode. In step **S0**, it is assumed that the work vehicle **1** is set to the second operation mode and the operation of the traveling machine body **11** is stopped.

[0099] In step **S1**, the traveling controller **113c** and the work machine controller **113d** determine whether an instruction for operating the traveling machine body **11** has been issued by the remote operation device **200**. In this embodiment, as described above, the operation instruction is realized by operating at least one of the first operation lever **204a**, the second operation lever **204b**, the first operation switch **205a**, and the second operation switch **205b** of the remote operation device **200**. For example, the forward movement instruction of the traveling machine body **11** is realized by tilting the first operation lever **204a** in the F1 direction. Furthermore, an instruction for lifting the work machine **12** (an instruction for driving the work machine lifting device **111e**) is realized by tilting the second operation lever **204b** in the F1 direction. Furthermore, by tilting the second operation lever **204b** in the F1 direction while tilting the first operation lever **204a** in the F1 direction, an instruction for moving the traveling machine body **11** forward and an instruction for lifting the work machine **12** are realized.

[0100] When an operation instruction has been issued (Yes in step **S1**), the process proceeds to next step **S2**. When an operation instruction has not been issued (No in step **S1**), the traveling controller **113c** and the work machine controller **113d** continue to determine the operation instruction.

[0101] In step **S2**, when the operation instruction is a travel instruction (a forward movement instruction, a backward movement instruction, a right turn instruction, or a left turn instruction), the traveling controller **113c** controls the traveling drive device **111c** to stop the operation of the traveling machine body **11** (keep the traveling machine body **11** stopped). For example, when a forward movement instruction of the traveling machine body **11** is issued (an operation of the first operation lever **204a** in the F1 direction), the traveling controller **113c** controls the electric motor of the traveling drive device **111c** against the forward movement instruction so that a rotational speed of an output of the electric motor becomes zero. Accordingly, the traveling of the traveling machine body **11** continues to be stopped.

[0102] Furthermore, when a backward movement instruction (operation of the first operation lever **204a** in the B1 direction), a left turn instruction (operation of the first operation lever **204a** and the second operation lever **204b** in the L1 direction), and a right turn instruction (operation of the first operation lever **204a** and the second operation lever **204b** in the R1 direction) of the traveling machine body **11** are issued, the traveling of the traveling machine body **11** continues to be stopped similarly to the case of the forward movement instruction. Note that the same applies to a case where an automatic traveling start instruction (pressing of the first operation switch **205a**) is issued as the case where forward movement instruction is issued.

[0103] Furthermore, when the operation instruction is an instruction for lifting or lowering the work machine **12**, the work machine controller **113d** controls the work machine lifting device **111e** to stop (continue to stop) the operation of the traveling machine body **11**. For example, when an instruction for lifting the work machine **12** (operation of the second operation lever **204b** in the F1 direction) is issued, the work machine controller **113d** controls at least one of the hydraulic pump **111e1** and the directional control valve **111e2** so that supply of hydraulic oil to the lifting cylinder **111e3** is blocked, contrary to the lifting instruction. More specifically, at least one of stopping of

driving of the hydraulic pump **111e1** and blocking of a flow path by the directional control valve **111e2** is performed. As a result, in the traveling machine body **11**, driving of the work machine lifting device **111e**, that is, lifting and lowering of the work machine **12**, continues to be stopped. Note that, also in a case where a lowering instruction of the work machine **12** is issued (operation of the second operation lever **204b** in the B1 direction), similarly to a case where a lifting instruction is issued, in the traveling machine body **11**, the driving of the work machine lifting device **111e** continues to be stopped.

[0104] Furthermore, when the operation instruction is an instruction for driving the work machine **12**, the work machine controller **113d** controls the work machine drive device **111d** to stop the operation of the traveling machine body **11** (keep the traveling machine body **11** stopped). For example, when a drive instruction for the work machine **12** is issued (pressing of the second operation switch **205b**), the work machine controller **113d** controls the electric motor of the work machine drive device **111d** against the drive instruction so that the rotational speed of the electric motor becomes zero. As a result, in the traveling machine body **11**, the driving of the work machine drive device **111d**, that is, the driving of the work machine **12**, continues to be stopped.

[0105] When the work vehicle **1** is set to the second operation mode, the operator is often positioned near the in-vehicle operation device **111b**, that is, near the traveling machine body **11**, in order to operate the traveling machine body **11** using the in-vehicle operation device **111b**. In this case, for example, when the operation of the traveling machine body **11** by the remote operation device **200** is changed (returned) from a disabled state to an enabled state and a worker other than the operator erroneously operates the remote operation device **200**, the traveling machine body **11** and the operator may come into contact with each other. In addition, the operator may come into contact with not only the traveling machine body **11** itself but also the work machine **12** that operates (drives, moves up and down) in conjunction with the operation of the traveling machine body **11**. Therefore, the following configuration is desirable from the viewpoint of ensuring the safety of the work vehicle **1** in the second operation mode even when the operation of the traveling machine body **11** by the remote operation device **200** is changed from the disabled state to the enabled state. That is, as in this embodiment, it is desirable to prohibit the traveling machine body **11** from being operated by using the remote operation device **200** in the second operation mode.

[0106] Note that, in the first operation mode, the traveling machine body **11** is prohibited from being operated using the in-vehicle operation device **111b**, but may be allowed to be operated.

[0107] Here, speed limitation of the traveling machine body **11** in the second operation mode will be described. Hereinafter, as an example of the speed limitation of the traveling machine body **11** in the second operation mode, a case where the traveling machine body **11** moves forward will be described. However, the same applies to cases where the traveling machine body **11** moves backward and turns (turns to the left and turns to the right).

[0108] FIG. **11** is a graph illustrating a change in a target speed V_t of the traveling machine body **11** in the second operation mode with respect to an operation amount X of the forward movement operation button OB-F. Note that FIG. **11** also illustrates a change in the target speed V_t of the traveling machine body **11** in the first operation mode with respect to an operation amount Y of the first operation lever **204a** in the F1 direction. In FIG. **11**, a solid line indicates a change in the target speed V_t of the traveling machine body **11** with respect to the operation amount X , and a broken line indicates a change in the target speed V_t of the traveling machine body **11** with respect to the operation amount Y .

[0109] Note that the operation amount X of the forward movement operation button OB-F (see FIG. **9**) means a pressing amount from a neutral state (non-operation state). For example, when the forward movement operation button OB-F is completely pressed (fully pressed), the operation amount X is 100%. Furthermore, the operation amount Y of the first operation lever **204a** means a tilting amount from the neutral state (non-operation state). For example, the operation amount Y is 100% when the first operation lever **204a** is completely tilted (fully tilted) in the F1 direction.

Furthermore, the target speed V_t means a target speed when the traveling controller **113c** controls the traveling drive device **111c** to cause the traveling machine body **11** to travel. That is, the traveling controller **113c** controls the traveling drive device **111c** so that an (actual) speed V of the traveling machine body **11** coincides with the target speed V_t . Therefore, the target speed V_t corresponds to the (actual) speed V of the traveling machine body **11**.

[0110] In the first operation mode, the target speed V_t of the traveling machine body **11** changes with the operation amount Y of the first operation lever **204a** (see the broken line in FIG. **11**). More specifically, when the operation amount Y increases, the target speed V_t also increases, that is, the speed V of the traveling machine body **11** also increases. For example, when the operation amount Y is 100% (when the first operation lever **204a** is fully tilted in the F1 direction), the target speed V_t becomes a maximum speed V_{max} (for example, 15 km/h), and the traveling machine body **11** travels at the maximum speed V_{max} . Note that, as described above, the maximum speed V_{max} can be adjusted by the operation knobs **206** (see FIG. **3**) of the remote operation device **200**. Here, in the first operation mode, a range of the speed V in which the traveling machine body **11** can travel is referred to as a speed range SR . In addition, in the speed range SR , a range exceeding a speed threshold value V_a (for example, 5 km/h) is referred to as a high speed range $SR-H$, and a range equal to or less than the speed threshold value V_a is referred to as a low speed range $SR-L$.

[0111] In the second operation mode, the target speed V_t of the traveling machine body **11** is constant at a speed limit V_{lim} (for example, 3 km/h) when the operation amount X of the forward movement operation button **OB-F** is equal to or greater than an operation amount threshold value X_a (for example, 90%) (see the solid line in FIG. **11**). For example, when the forward movement operation button **OB-F** is completely pressed, the target speed V_t becomes the speed limit V_{lim} , and the traveling machine body **11** travels at the speed limit V_{lim} . The speed limit V_{lim} is, for example, a preset fixed value. More specifically, the speed limit V_{lim} is set within the range of the low speed range $SR-L$ (that is, less than the speed threshold value V_a), and is stored in the storage **113a**. That is, in the second operation mode, the speed V of the traveling machine body **11** is limited to the low speed range $SR-L$.

[0112] Note that the change in the target speed V_t of the traveling machine body **11** in the second operation mode with respect to the operation amount X is not limited to the above. For example, the target speed V_t of the traveling machine body **11** in the second operation mode may change together with the operation amount X . However, the target speed V_t preferably changes within the low speed range $SR-L$. Furthermore, the operation amount threshold value X_a is not limited to 90%, and may be, for example, 80% or 100%. Furthermore, the speed limit V_{lim} is not limited to a fixed value and may be, for example, a variable value. That is, the speed limit V_{lim} may be adjustable similarly to the maximum speed V_{max} . However, the speed limit V_{lim} is preferably adjustable within the low speed range $SR-L$.

[0113] The speed V of the traveling machine body **11** in the case where the speed V of the traveling machine body **11** in the second operation mode is set within the low speed range $SR-L$ is lower than the speed V of the traveling machine body **11** in the case where the speed V is set within the high speed range $SR-H$. Thus, for example, even when the traveling machine body **11** approaches, the operator can easily perform an operation for avoiding a collision with the traveling machine body **11**. Therefore, from the viewpoint of improving safety during movement of the traveling machine body **11** in the second operation mode, the following configuration is desirable. That is, as in this embodiment, in the second operation mode, the speed V of the traveling machine body **11** is desirably limited to the low speed range $SR-L$ in the speed range SR of the traveling machine body **11** in the first operation mode.

6. OPERATION MODE SWITCHING SETTING

[0114] A setting of switching between the first operation mode and the second operation mode will be described with reference to FIG. **12**. FIG. **12** is a flowchart illustrating a flow of a setting of switching between the operation modes. It is assumed that the first operation mode is set as the

operation mode in step S10.

[0115] In step S11, the operation mode switcher **113b** (see FIG. 7) determines whether an operation mode switching instruction has been issued. In this embodiment, the switching instruction is realized by the operator operating (touching) the second operation mode setting icon IC2 (see FIG. 6) of the switching manipulator **111b4** with a finger or the like. When the switching instruction has been issued (Yes in step S11), the processing proceeds to next step S12. When the switching instruction has not been issued (No in step S11), the operation mode switcher **113b** continues the determination of the switching instruction.

[0116] In step S12, the operation mode switcher **113b** switches the operation mode from the first operation mode to the second operation mode. Thus, the operation mode is set to the second operation mode. That is, the switching manipulator **111b4** enables switching from the first operation mode to the second operation mode. As described above, when the operation mode is set to the second operation mode, the second operation mode display icon IC2a of the switching manipulator **111b4** is turned on (or blinked) (see FIG. 6).

[0117] For example, the following configuration is desirable from the viewpoint of reliably performing switching from the first operation mode to the second operation mode even when a failure, power shortage, or the like occurs in the remote operation device **200**. That is, as illustrated in FIG. 4, in a configuration in which the work vehicle **1** includes the switching manipulator **111b4** that enables switching from the first operation mode to the second operation mode, it is desirable that the switching manipulator **111b4** be provided in the traveling machine body **11**.

[0118] Note that the switching instruction (from the first operation mode to the second operation mode) is not limited to the above-described configuration. For example, the switching instruction may be realized by the operator continuously pressing the forward movement operation button OB-F of the traveling manipulator **111b2** for a predetermined period of time (for example, one second) instead of (or in addition to) the operation of the switching manipulator **111b4** (particularly, the second operation mode setting icon IC2). That is, the traveling manipulator **111b2** may be capable of issuing a traveling instruction to the traveling machine body **11** and performing switching from the first operation mode to the second operation mode.

[0119] In addition, for example, the switching instruction may be realized by the operator continuously pressing the operation button OB other than the forward movement operation button OB-F included in the traveling manipulator **111b2** for a predetermined period of time. That is, the switching instruction may be realized by the operator continuously pressing any one of the backward movement operation button OB-B, the left turning operation button OB-L, and the right turning operation button OB-R for a predetermined period of time. Alternatively, the switching instruction may be realized by pressing at least two of the forward movement operation button OB-F, the backward movement operation button OB-B, the left turning operation button OB-L, and the right turning operation button OB-R together, or by continuously pressing them for a predetermined period of time.

[0120] In order to reduce the burden of the setting work on the operator, it is desirable to reduce the number of manipulators for the setting work (setting operation) in the work vehicle **1**. From this viewpoint, as in this modification, in the configuration in which the in-vehicle operation device **111b** includes the traveling manipulator **111b2**, it is desirable that the traveling manipulator **111b2** enables switching from the first operation mode to the second operation mode.

[0121] Note that, in step S10, the operation mode may be set to the second operation mode. In this case, the operation mode is switched from the second operation mode to the first operation mode by the switching setting described above. The switching instruction in this case is realized by operating the first operation mode setting icon IC1 (see FIG. 5) of the switching manipulator **111b4**. That is, the switching manipulator **111b4** enables switching from the second operation mode to the first operation mode in addition to the switching from the first operation mode to the second operation mode.

7. NOTIFICATION TIMING OF NOTIFIER

[0122] A timing at which the notifier **111f** (see FIGS. 1 and 7) performs notification will be described with reference to FIG. 13. FIG. 13 is a flowchart of a flow when the notifier **111f** performs notification. It is assumed that the second operation mode is set as the operation mode in step S20.

[0123] In step S21, the notification controller **113e** (see FIG. 7) determines whether a forward movement instruction for the traveling machine body **11** has been issued by the in-vehicle operation device **111b**. As described above, in this embodiment, the forward movement instruction is realized by the operator operating the forward movement operation button OB-F of the in-vehicle operation device **111b**. When the forward movement instruction has been issued (Yes in step S21), the process proceeds to next step S22. When the forward movement instruction has not been issued (No in step S21), the notification controller **113e** continues to determine whether the forward movement instruction has been issued.

[0124] In step S22, the notification controller **113e** controls the notifier **111f** (speaker) to make a notification (sound). That is, in the second operation mode, the notifier **111f** makes a notification when the traveling machine body **11** moves to one side in one direction (in this embodiment, the front side in the front-back direction).

[0125] When the work vehicle **1** is set to the second operation mode in a configuration in which the in-vehicle operation device **111b** is disposed on one side (the front side in the front-back direction in this embodiment) in one direction included in the horizontal direction in the traveling machine body **11**, the operator is often positioned as follows. That is, the operator is often positioned on one side in one direction (in this embodiment, the front side in the front-back direction) with respect to the traveling machine body **11** in order to operate the traveling machine body **11** by using the in-vehicle operation device **111b**. In this case, for example, when the operator erroneously (unintentionally) moves the traveling machine body **11** to one side in one direction (moves forward in this embodiment), the traveling machine body **11** may approach the operator and come into contact with the operator. Therefore, even when the operator makes a mistake in operation, it is desirable to make the operator recognize the approach of the traveling machine body **11** in order to avoid contact between the operator and the traveling machine body **11**. From this point of view, as in this embodiment, it is desirable that the notifier **111f** performs notification when the traveling machine body **11** moves to one side in one direction in the second operation mode.

8. APPENDICES

[0126] The work vehicle **1** described in this embodiment can also be expressed as a work vehicle described in the following appendices.

[0127] A work vehicle according to Appendix 1, comprising: [0128] a traveling machine body; [0129] an in-vehicle operation device that is installed in the traveling machine body and that operates the traveling machine body; and [0130] a remote operation device that is provided separately from the traveling machine body and that operates the traveling machine body, wherein [0131] the work vehicle has [0132] a first operation mode for operating the traveling machine body using the remote operation device, and [0133] a second operation mode for operating the traveling machine body using the in-vehicle operation device.

[0134] A work vehicle according to Appendix 2, wherein, in the work vehicle according to Appendix 1, [0135] in the second operation mode, the traveling machine body is prohibited from being operated using the remote operation device.

[0136] A work vehicle according to Appendix 3, in the work vehicle according to Appendix 1 or 2, further comprising: [0137] a switching manipulator capable of performing switching from the first operation mode to the second operation mode, wherein [0138] the switching manipulator is disposed in the traveling machine body.

[0139] A work vehicle according to Appendix 4, in the work vehicle according to Appendix 1 or 2, the in-vehicle operation device including: [0140] a traveling manipulator capable of issuing a

traveling instruction to the traveling machine body, wherein [0141] the traveling manipulator is capable of performing switching from the first operation mode to the second operation mode. [0142] A work vehicle according to Appendix 5, in the work vehicle according to any one of Appendices 1 to 4, further comprising: [0143] a notifier, wherein [0144] the in-vehicle operation device is disposed on one side in one direction included in a horizontal direction in the traveling machine body, and [0145] the notifier makes a notification when the traveling machine body moves to the one side in the one direction in the second operation mode.

[0146] A work vehicle according to Appendix 6, wherein, in the work vehicle according to any one of Appendices 1 to 5, [0147] in the second operation mode, a speed of the traveling machine body is limited to a low speed range in a speed region of the traveling machine body in the first operation mode.

[0148] The embodiments of the present invention have been described above, but the scope of the present invention is not limited thereto, and may be expanded or modified without departing from the gist of the invention.

INDUSTRIAL APPLICABILITY

[0149] The present invention is applicable to work vehicles, such as an agricultural machine and a construction machine.

REFERENCE SIGNS LIST

[0150] **1** work vehicle [0151] **11** traveling machine body [0152] **111b** in-vehicle operation device [0153] **111b2** traveling manipulator [0154] **111b4** switching manipulator [0155] **111f** notifier [0156] **200** remote operation device [0157] V speed (speed of traveling machine body) [0158] SR speed range [0159] SR-L low speed range

Claims

1. A work vehicle, comprising: a traveling machine body; an in-vehicle operation device that is installed in the traveling machine body and that operates the traveling machine body; and a remote operation device that is provided separately from the traveling machine body and that operates the traveling machine body, wherein the work vehicle has: a first operation mode for operating the traveling machine body using the remote operation device, and a second operation mode for operating the traveling machine body using the in-vehicle operation device.
 2. The work vehicle according to claim 1, wherein in the second operation mode, the traveling machine body is prohibited from being operated using the remote operation device.
 3. The work vehicle according to claim 1, further comprising: a switching manipulator capable of performing switching from the first operation mode to the second operation mode, wherein the switching manipulator is disposed in the traveling machine body.
 4. The work vehicle according to claim 1, the in-vehicle operation device including: a traveling manipulator capable of issuing a traveling instruction to the traveling machine body, wherein the traveling manipulator is capable of performing switching from the first operation mode to the second operation mode.
 5. The work vehicle according to claim 1, further comprising: a notifier, wherein: the in-vehicle operation device is disposed on one side in one direction included in a horizontal direction in the traveling machine body, and the notifier makes a notification when the traveling machine body moves to the one side in the one direction in the second operation mode.
 6. The work vehicle according to claim 1, wherein, in the second operation mode, a speed of the traveling machine body is limited to a low speed range in a speed region of the traveling machine body in the first operation mode.
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